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Categorization of Colorectal Polyps using Convolutional Neural Networks for Enhanced Diagnostic Accuracy and Early Detection in Colorectal Cancer Screening

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Abstract*— Colon cancer is a serious health risk that increases human disease and death. Histological analysis is necessary for a definitive diagnosis of this illness. A promising approach to the study of colon cancer is the integration of convolutional neural networks (CNNs) in digital image processing, especially with whole-slide imaging. Accurately classifying colon adenocarcinomas is essential for efficient testing. Creating a system that uses image analysis to recognize and classify colon adenocarcinomas by utilizing certain preparatory techniques to develop a unique computer program that use a Convolutional Neural Network (CNN) to enhance and analyze digital pathology images. By using machine intelligence to automate the diagnosis of cancer, more people can be evaluated more quickly and affordably, improving the effectiveness and accessibility of healthcare evaluations. This automation is made possible by the use of state-of-the-art methods for complex digital image manipulation and advanced pattern identification in data. The findings show that the suggested framework has an astounding accuracy rate of up to 99% when it comes to the careful analysis of cancer tissues. By enhancing the model's generalization over bigger datasets and evaluating its resilience in diverse clinical settings, we hope to advance this research. By using this technique, medical professionals will be able to develop a trustworthy automated system for identifying various forms of colon cancer.

Keywords*— Deep learning, Cancer Detection, Convolutional Neural Network, Colorectal cancer, Machine Learning Introduction

I. INTRODUCTION

As per the WHO, cancer is the primary cause of mortality worldwide. In particular, colorectal cancer, which starts in the colon or rectum, belongs to a larger group of illnesses that are marked by the spread of faulty cells as a result of random mutations, which causes the organs to develop out of control. With nearly 10 million deaths from cancer in 2020, it ranks as the second leading reason for passing away worldwide, after cardiovascular illnesses.

With about 940,000 cases and over 500,000 fatalities yearly, colon and rectal cancers rank second in developed societies. These malignancies are on the rise worldwide, affecting people of all ages, but particularly those in their 50s and 60s, although being less common in less developed regions. Forecasts indicate a potential 60%.

The development of non-cancerous polyps on the colon's inner wall is usually the first sign of colon cancer. Tumors may develop from some of these polyps if they become malignancy. One common kind of colon cancer that arises from the cells lining the large intestine is called an adenocarcinoma. Aggressive variants include ring cell adenocarcinomas and mucinous adenocarcinomas. Physical traits can change over time depending on a variety of circumstances, including age, sex, and smoking habits. A few genetic diseases can cause changes in a comparatively short amount of time.

Regular screenings are essential for people who are at risk of genetic cancers, but the expense of diagnostic tests can be prohibitive, especially in low-income countries where about 70% of cancer deaths occur. Such nations have limited access to pathology support for cancer detection, which highlights the urgent need for large investments in laboratories, skilled personnel, and public health infrastructure. It takes a lot of money and effort to solve these problems, which stimulates research into other diagnostic modalities. Machine learning (ML) in computer science appears to be a promising field with potential applications in pathology and automated cancer detection. Pathologists' traditional method of histopathological image processing requires a lot of resources, which has led to a move towards ML-based computer-aided diagnosis systems. Neural network-based learning algorithms, particularly those for image recognition, have demonstrated efficacy across a range of applications. The research use a deep learning lens to analyze colon cancer using histology and a suggested deep convolutional neural network (DCNN) model. This model is highly efficient in learning characteristics without the need for manual filter tweaks, and it is inspired by the way the human brain functions. It requires minimal setup before use. CNNs are used in screening because, for example, adenocarcinomas frequently exhibit distinctive morphological features on histological slides, such as glandular structures.

II. LITERATURE REVIEW

The promise of this subject was originally recognized in 1955 when Lee Lusted first envisioned the use of computers in medical diagnosis [7]. This early realization paved the way for the application of computational approaches in healthcare. When the practical use of computers in medical picture diagnosis was established in 1963, diagnostic assistance technologies saw a radical shift [8]. Histopathological slides can now be converted into digital images thanks to the development of affordable digital scanners, which has made data exchange, analysis, and archiving easier [6].

The introduction of deep learning (DL) techniques, which increased automation and precision, drastically altered medical diagnoses. Lung, breast, and colon cancer have all been diagnosed using these techniques, which offer valuable insights into a range of malignancies [9, 10]. For example, in their 2013 proposal of a multimodal sparse features classification (mSRC) approach for lung cancer diagnosis, Shie et al. [11] enhanced diagnostic performance by combining many data sources. In 2014, Jin [12] introduced an automated diagnostic support (CAD) method that used CT scan data to classify lung cancer with an average accuracy of 88.10%.

In order to classify histological colon images in the field of colon cancer detection, Xu and colleagues employed Support Vector Machines (SVMs) to extract particular features from the data [13]. As deep learning gained popularity, convolutional neural networks (CNNs) became the preferred technique for diagnosing cancer. In order to identify cancer in whole-slide images of colon tissues, a study by ResNet-based researchers [14] examined three CNN architectures (ResNet-18, ResNet-30, and ResNet-50). The potential of CNNs in histopathological analysis was demonstrated by ResNet-50, which surpassed the other models with an accuracy of 93.91%.

In addition to CNNs, researchers have focused on addressing the challenges of classifying histopathology images. A study [15] that trained CNNs for colon cell image categorization [16] using a multi-step training approach showed the value of iterative refining techniques. Meanwhile, Toraman et al. looked into the detection of colon cancer using Fourier Transform Infrared (FTIR) spectroscopy. Their research incorporated visualization techniques like GradCAM and SmoothGrad to increase interpretability through the identification of cancerous regions in pre-trained CNN models.

Among other aspects of histology, deep learning applications have grown to encompass the classification of cells and nuclei. Dale et al. [17] used these techniques, for instance, to assess nuclear pleomorphisms in breast cancer images. Malon and Cosatto [18] trained CNNs to distinguish between mitotic and non-mitotic cells using color, texture, and shape-based signals. Other publications [19–21] looked into the segmentation and classification of nuclei utilizing appearance, area, and shape feature analysis. Comparative research, such as that done by Abbas et al. [22], evaluated how well a number of CNN designs detected squamous cell carcinomas.

In the context of colon cancer, Wang et al. [23] proposed a two-step classification model for colonic adenocarcinomas that integrates manually generated features with CNN discovered one. The benefits of traditional feature engineering and deep learning-based feature extraction are combined in these hybrid approaches. Studies by Bukhari et al. [24] looked at various ResNet architectures to identify variations in model performance, showing how architecture selection influences diagnostic outcomes. Masud et al. [24] used lightweight CNNs to detect lung nodules from CT scans, achieving good accuracy and minimal processing costs. Shakeel et al. [25] shown the adaptability of computer-aided cancer detection methods in a range of clinical contexts by presenting a unique lung cancer screening strategy using CT scan-based CAD systems.

Though they struggle with high-dimensional datasets, traditional machine learning methods like Support Vector Machines (SVM) have demonstrated value in the diagnosis of cancer. However, because CNNs can automatically learn features, they are more appropriate for histopathology applications. However, issues like overfitting require the use of advanced techniques like data augmentation

III. METHODOLOGY

Figure 1 illustrates the many steps of our inquiry, which are defined in this section. Figure 1 displays a flow diagram of the suggested work. Every step is covered in detail, including feature extraction, training, data preprocessing, model evaluation, and prediction. Undertaking a classification job that differentiates between benign and cancer tissue using a convolutional neural network (CNN) and knowledge transfer models is the main objective of this phase. To enhance the performance, we suggest a CNN model that incorporates additional depth. Furthermore, using the same dataset, we perform a comparative analysis of other transfer learning models, such as ResNet50, DenseNet121, ResNet101 V2, Efficient NetB0, VGG-16, and MobileNetV2.

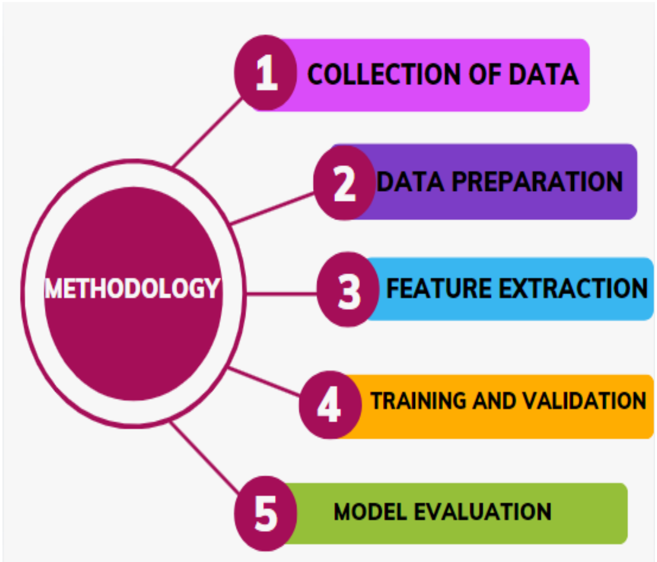

Fig 1: Methodology

A. Data Set Details

The dataset includes 10,000 digital photos that show histopathology slides. H&E-stained visual depictions of osteosarcoma tissues. Table 1 provides a comprehensive summary of the dataset, comprising 250 photos of benign (non-cancerous) and colon adenocarcinomas (a type of colon cancer) colon tissue. It is important to know that colon adenocarcinoma is the most common kind of colon cancer, making up approximately 95% of cases. This particular sort of polyp, known as an adenoma, arises in the colon and develops into malignant tissue, which is the source of the cancer.

NEOPLASM	CLASS LABEL	SAMPLE COUNT
Colon adenocarcinoma	Colon_aca	5000
Colon benign tissue	Colon_n	5000

Table 1: Data Set Details

B. Data Preprocessing

Our preprocessing phase's main objective is to produce images appropriate for the detection system's later stages. Data pretreatment is the first crucial stage in preparing data for ML model integration. In our suggested investigation, image diagnosing and artefact correction were necessary to get a high classification rate. Reduce data, formalize data, extract feature, and convert string data to numeric data were the preparation processes we took. Reduction, which involves mapping a dimensional from high to a lower dimensional space that contains more important information, is an essential step before model building.

C. Data Normalization

A technique called data normalization is used to organize a dataset so as to reduce biases, outliers, updates, and removal discrepancies, among other undesirable qualities. Normalization can be accomplished in a number of ways, including min-max, z-score, and decimal scale. normalization. In our investigation, we employed z-score normalization to make sure that our dataset was consistent. This procedure includes both the theoretical comprehension and the actual application of putting various variables on a uniform scale. To make it easier to compare scores across different kinds of criteria is the goal. The fundamental idea is to convert the data into a common scale by standardizing it, where standardized 65 normalized to a standard scale. Z-score provide a standardized reference point by indicating the number of standard deviations above or below the mean.

D. Feature Extraction

A crucial stage in image processing is feature extraction, which divides an image into smaller, easier-to-manage components for additional examination. In our work, we extract features from enormous datasets in order to better detect and identify trends. This entails transforming the input data (pictures) into a collection of unique, non- repetitive traits. These characteristics are descriptive variables that facilitate the machine learning process during the learning and observation stages.

In our study, we employed PCA to identify features within our collection of photos. When it comes to minimizing information loss and lowering dimensionality, PCA is especially useful. The most important sources of variance in the data are the principal components that PCA was able to extract. PCA maximizes under specific signal and noise models. The goal of PCA is to maximize the mutual information between the reduced-depth output and the required data. The following elements are involved in the PCA process:

1. **Data Normalization:**

Prior to doing PCA, it is essential to ensure consistency in the scale of unscaled data using various measurement units to avoid biases in the relative variance comparison across features. In order to minimize dimensionality and emphasize important changes, PCA was utilized for feature extraction. This method guarantees that the model concentrates on significant patterns found in the data.

2. **Eigen Decomposition Covariance Matrix Creation:**

Reducing the feature set requires understanding the overall percentage of variation collected by each key component in order to identify potential relationships between many parameters.

3. **Selecting the Ideal Amountof Principal Components:**

Analyzing the relationship between the number of major components can help determine the right number of components. An information loss trade-off must be made between dimensionality reduction and principal component selection for additional analysis.

4. **Transform into Numerical Information:**

Numerical numbers are the most often used input type in machine learning algorithms. We put in place a procedure to turn our data into numerical numbers, making sure that every property has a distinct scale. These data also need to be standardized and normalized in order to enhance the model validation and training processes for various deep learning (DL) control systems. In our experiment, we make use of the Label Encoder, a tool from the Python standard library. This tool assists us in converting the labels "adenocarcinoma" and "benign" to numerical values, denoted by 0 and 1.

E. Proposed CNN Architecture

Utilizing prior knowledge from an activity to inform a new undertaking is known as transfer learning. Transfer learning (TL) is adapting information from one task to another and becoming proficient in distant learning models. In essence, it's using data from a previously trained model to train new models on fresh input. TL is a useful tool for accelerating the development of artificial intelligence (AI) since it makes the process of fine-tuning a pre-trained model faster and easier.

It is difficult to train several neural network topologies on a large medical dataset like ImageNet. However, in our study, we trained multiple models, including our own CNN model and ResNet50, DenseNet121, Efficient NetB0, VGG-16, and Mobile Net V2, using a sizable medical dataset. Two categories in our proposed CNN model are used to distinguish between benign and cancerous colon tissue. We use the Adam optimizer to modify the weights and the backpropagation function to fine-tune the model. Furthermore, we use data augmentation to get around restrictions. We experimented with data augmentation techniques, such as rotation and flipping, to improve performance and produce more reliable models. Accuracy was further enhanced by hyper parameter optimization and transfer learning with ResNet50 and VGG-16.

IV. FINDINGS AND RESULTS

A. Splitting of Dataset

The picture shows a text output that validates the ML model loading and processing of a 10,000-image dataset. The training set, which consists of 8,000 pictures, is divided into two regions of the dataset in order to train the model. From these photos, the model will pick up patterns and characteristics that will increase its accuracy.

Test set: After training, the model's performance will be assessed using these 2,000 photos. On these images, the model will predict things, and its precision will be checked against the right responses. As part of a conventional machine learning procedure, the dataset is split into two as training and testing sets to prevent overfitting and ensure the model can adjust well to new data.

```
Loaded 10000 images.
Training set size: 8000
Test set size: 2000
```

Fig 2: Dataset Details used for Training and Testing.

B. Architecture of CNN Model

The picture displays a CNN model. The model is sequential, which means that its layers are built one on top of the other.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 224, 224, 16)	448
batch_normalization (BatchNormalization)	(None, 224, 224, 16)	64
max_pooling2d (MaxPooling2D)	(None, 112, 112, 16)	0
conv2d_1 (Conv2D)	(None, 112, 112, 32)	4,640
batch_normalization_1 (BatchNormalization)	(None, 112, 112, 32)	128
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 32)	0
conv2d_2 (Conv2D)	(None, 56, 56, 64)	18,496
batch_normalization_2 (BatchNormalization)	(None, 56, 56, 64)	256

Fig 3: CNN Architecture

C. Sample Dataset Images.

There are two possible categories:"colon_aca" or "colon_n". These photos most likely relate to colon tissue based on the labels. The two groups may correspond to distinct colon tissue types or pathologies.

"colon_aca" can be an acronym for adenocarcinoma, a prevalent form of colon cancer.

"colon_n" could be used to describe either healthy or disease-free colon tissue.

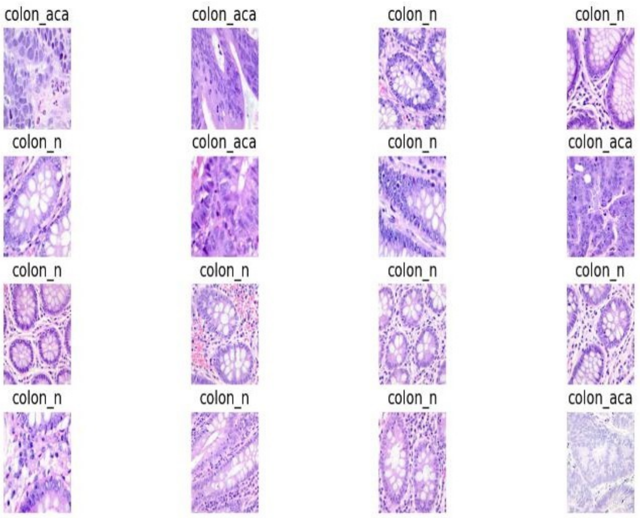

Fig 4: Dataset Images

D. Training and Validation plots of Loss and Accuracy

Loss on the left plot:

Blue Line Indicates the reduction in training. This is the loss that was determined for each epoch using the training set. As the model becomes more experienced, training loss frequently decreases, indicating that the model is becoming more adept at fitting training data.

The validation loss is represented by the purple line. This is the loss determined using a different, non-training validation dataset. An effective indicator for assessing the model's capacity to generalize to new data is the validation loss.

Plot Correct (Accuracy):

The blue line indicates the accuracy of training. This is the percentage of accurate forecasts made using the training set for every epoch. The training accuracy of the model typically rises with training, meaning that the model is using the training data to make predictions that are more accurate. Purple Line indicates the correctness of validation. This represents the percentage of accurate forecasts made using the validation data for every epoch. An effective statistic for assessing the model's capacity to generalize to new data is the validation accuracy.

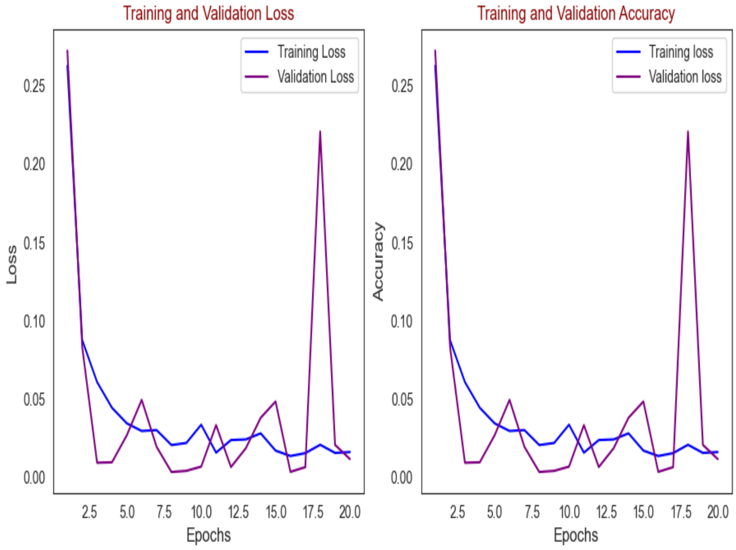

Fig 5: Loss and Accuracy Plots

E. Confusion Matrix Analysis

The confusion matrix, a visual aid for assessing a classification model's performance, is depicted in the image. Here, the confusion matrix pertains to a model that divides colon tissue into two groups: "colon_aca" and "colon_n".

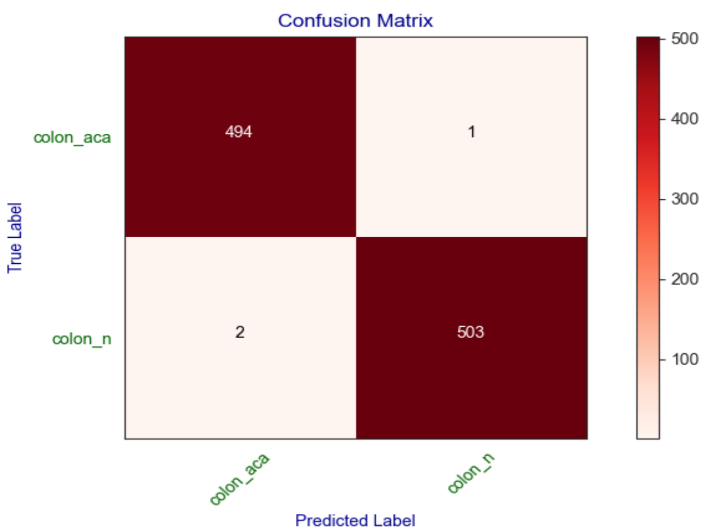

Fig 6: Confusion Matrix

F. Histogram Plot Representation

The image shows the training and validation accuracy histogram of a ML model. A histogram is a graphical representation of the distribution of a numerical collection. In this case, the y-axis displays the frequency of each accuracy value, while the x-axis displays the accuracy values. With a little tendency to overfit the training data, the histogram generally indicates that the model is performing well on both the training and validation data.

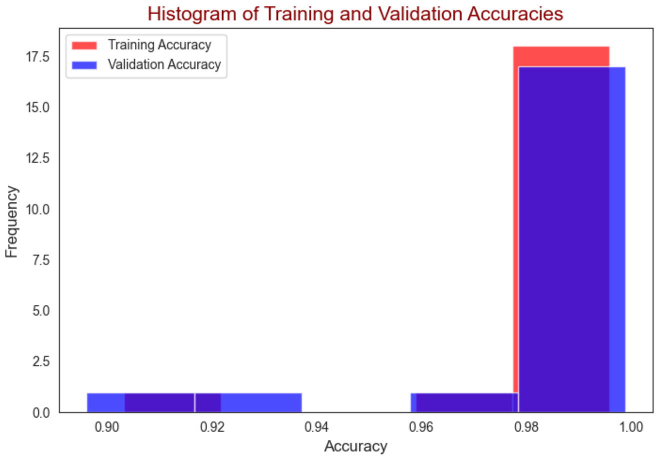

Fig 7: Histogram of Training and Validation Accuracies

G. Box plot Analysis of Training and Validation Losses:

The model generally achieved low loss on the training data, as seen by the comparatively low median training loss. Additionally, the IQR is not very high, indicating that the training losses were quite uniform. A small number of outliers on the higher end may point to some situations in which the model had difficulty fitting the training set.

The model performed marginally worse on the validation data than it did on the training data, according to the median validation loss, which is marginally higher than the training loss. Additionally, there is greater variability in the validation losses, as indicated by the bigger IQR for the validation loss. A few outliers are present on both the higher and lower ends, indicating that the validation data showed greater variability in the model's performance than the training data did. To guarantee strong performance, the evaluation approach computes precision, recall, and F1-scores. Additionally, we evaluated the model's dependability on unseen data and used cross-validation to avoid overfitting.

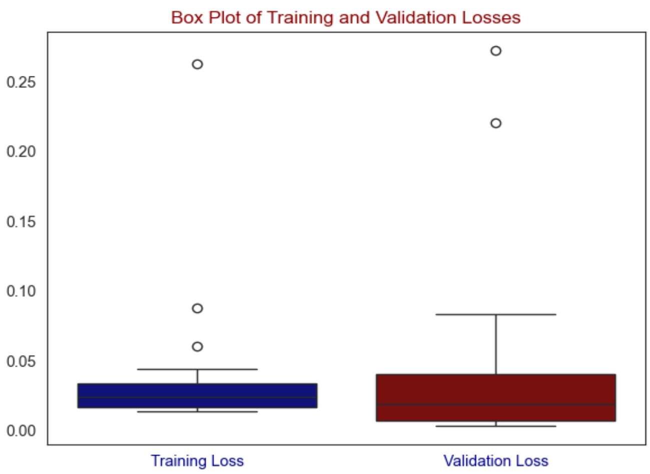

Fig 8: Box plot of Training and Validation Losses

H. Analysis Report

For both classes, the model's precision and recall were remarkably high, meaning that most of the positive cases were detected and accurate predictions were made.

The model's overall good performance is confirmed by the F1-score, which is also very high.

With an accuracy of 0.99, the model successfully identified 99% of the dataset's cases.

	precision	recall	f1-score	support
colon_aca	0.99	1.00	1.00	495
colon_n	1.00	0.99	1.00	505
accuracy			1.00	1000
macro avg	1.00	1.00	1.00	1000
weighted avg	1.00	1.00	1.00	1000

Fig 9: Analysis Report

V. FUTURE DIRECTIONS

Next objectives are to extend the study to a bigger dataset that contains more tagged samples of colon tissue. Developing a robust Deep Neural Network (DNN) with enhanced preprocessing methods is the aim in order to achieve optimal accuracy. In addition, we want to create a dependable and user-friendly Computer-Aided Diagnosis (CAD) system that can identify colon cancer in a range of imaging scenarios. Furthermore, our research route involves examining DNN models that can use CAD system to recognize various types of skin lesions. This feature aims to enhance the diagnostic skills of dermatology by rapidly and accurately identifying lesions through the use of deep learning.

VI. CONCLUSION

With 99% accuracy in differentiating between healthy and malignant tissues, the CNN-based approach to colorectal polyp classification exhibits great promise for improving diagnostic precision and early detection in colorectal cancer screening. This achievement shows how deep learning may automate histopathological analysis, enabling pathologists to make quicker and more precise diagnoses. Although there are still difficulties in improving the model for clinical usage, the combination of data augmentation and transfer learning improves the model's generalizability and lowers overfitting. Future studies will concentrate on growing the dataset, adding a variety of lesions, refining the architecture of the model, and creating a real-time computer-aided diagnostic (CAD) system. All things considered, this study shows how CNNs can revolutionize cancer diagnosis by increasing accessibility, dependability, and efficiency—particularly in environments with limited resources—and opening the door for all-encompassing AI-based diagnostic tools.

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